# JAUNDICE

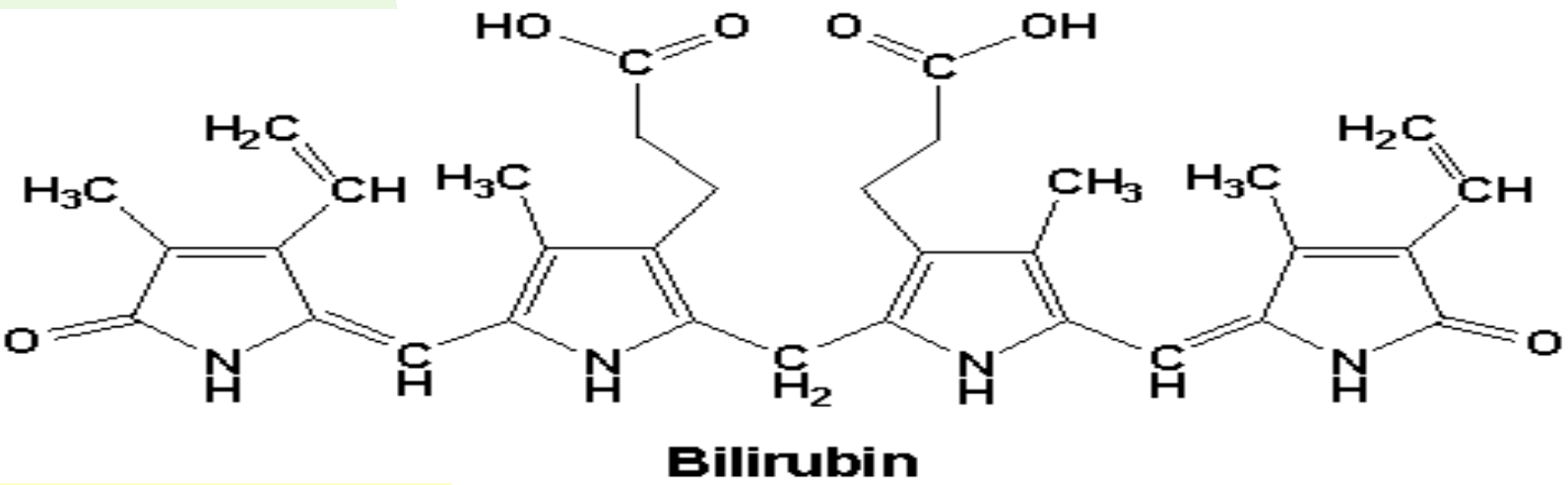
# JAUNDICE

- The word jaundice is from the French '*jaunisse*', meaning YELLOW
- Jaundice - also identified as ***icterus***.
- It is mainly due to yellowish or greenish pigmentation of the skin.
- Pigmentation appears usually eyes (white area) due to ***high bilirubin levels.***
- The color of feces - pale
- The color of urine - dark.

- Accumulation of bilirubin or its conjugates in body tissues produces jaundice (ie, ***icterus***), deposition bilirubin pigments in skin, sclera, mucous membranes, and other less visible tissues.
- At cellular level, when the normal functioning of the bilirubin metabolism and its excretion disturbs, jaundice may be the result.
- Levels of ***bilirubin*** in blood are normally below 1.0 mg/dL (17 µmol/L) whereas, levels over 2-3 mg/dL (34-51 µmol/L) results in jaundice.
- Causes of jaundice vary from non serious to potentially fatal.

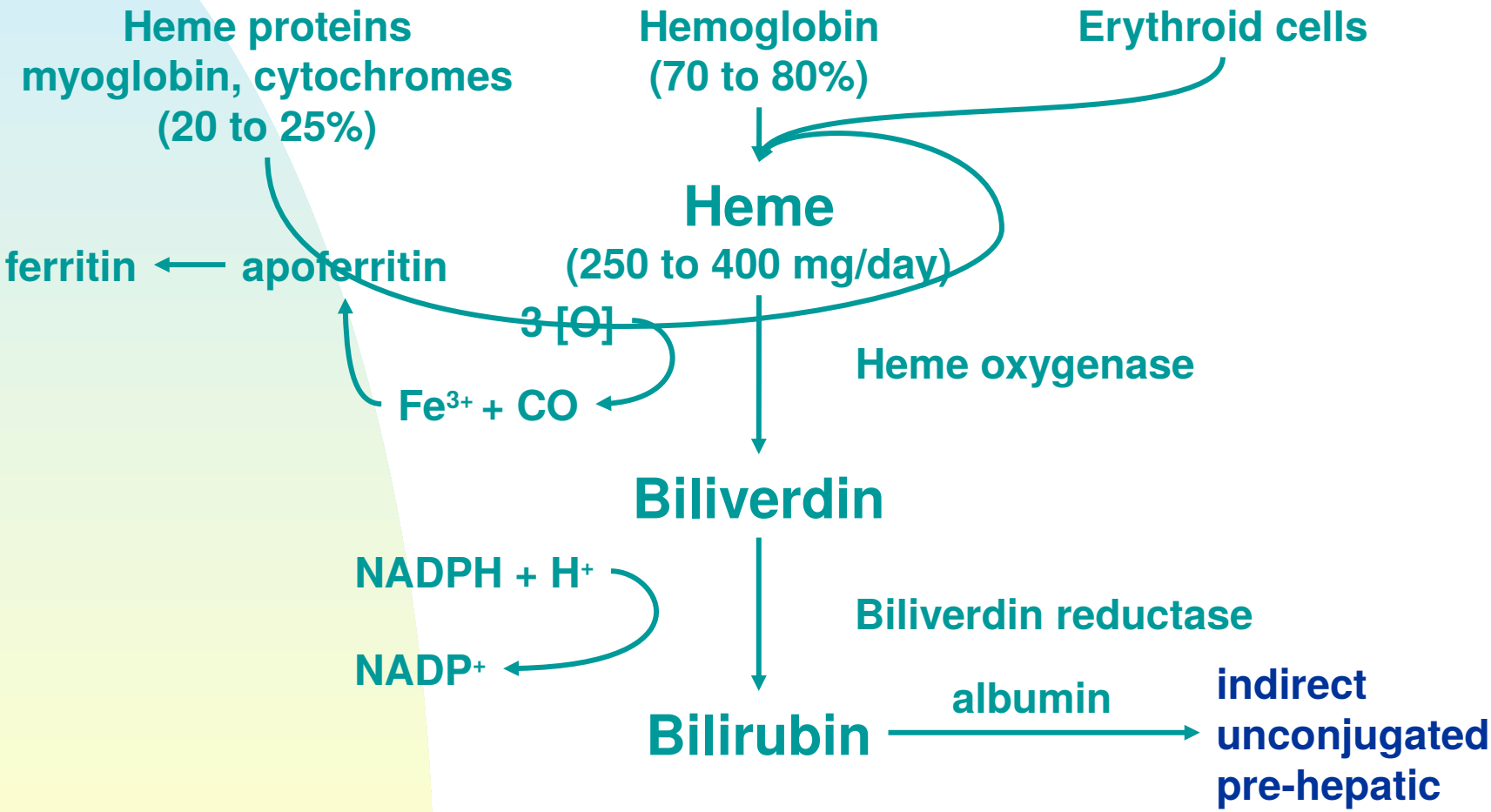
# What is Bilirubin?

- Breakdown product of heme metabolism
- Bile pigment and also excreted in urine.
- Lipophilic in nature (insoluble in aqueous medium).
- It consists of an open chain of four pyrrole-like rings (tetrapyrrole)

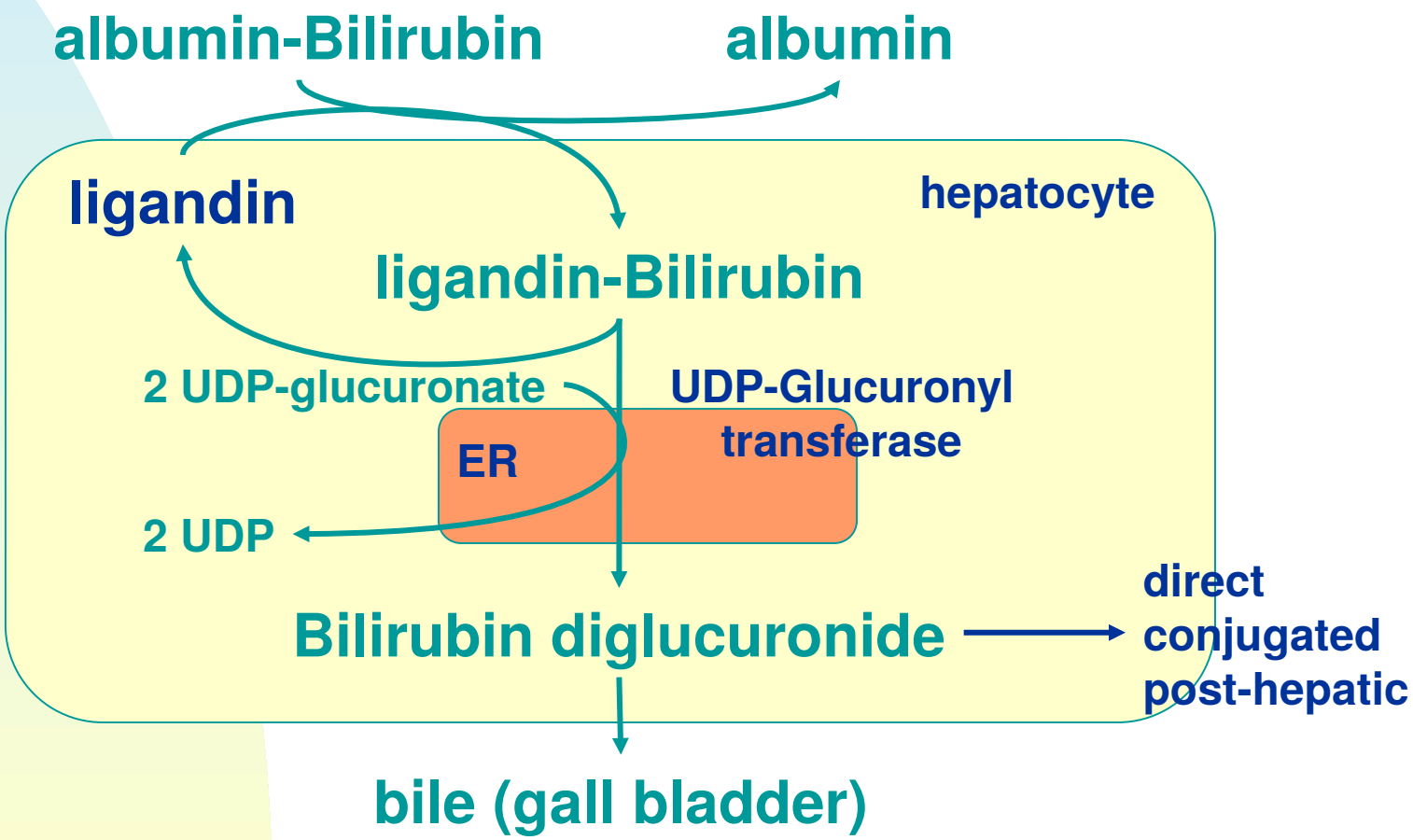



Bilirubin

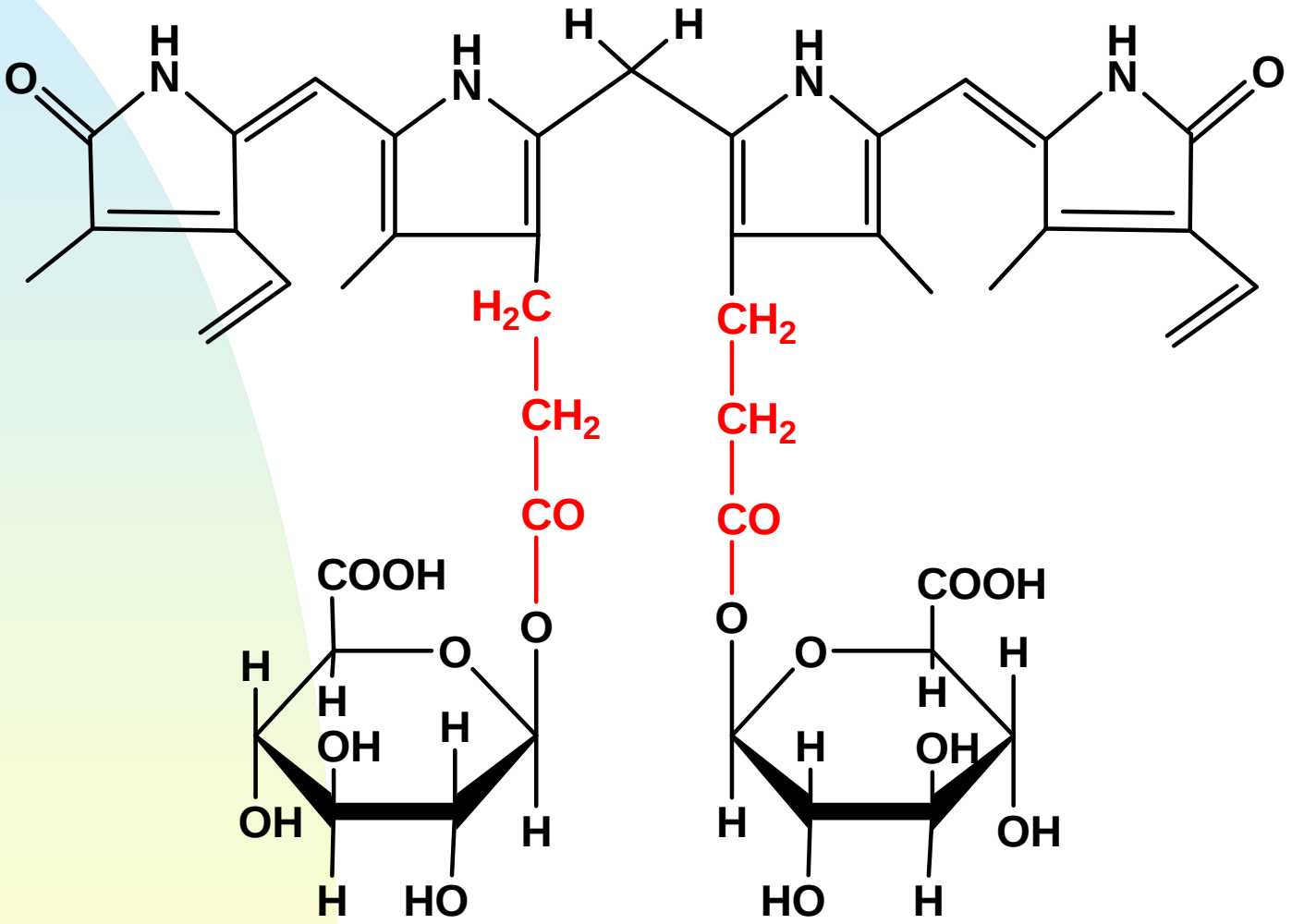
# Bilirubin Production



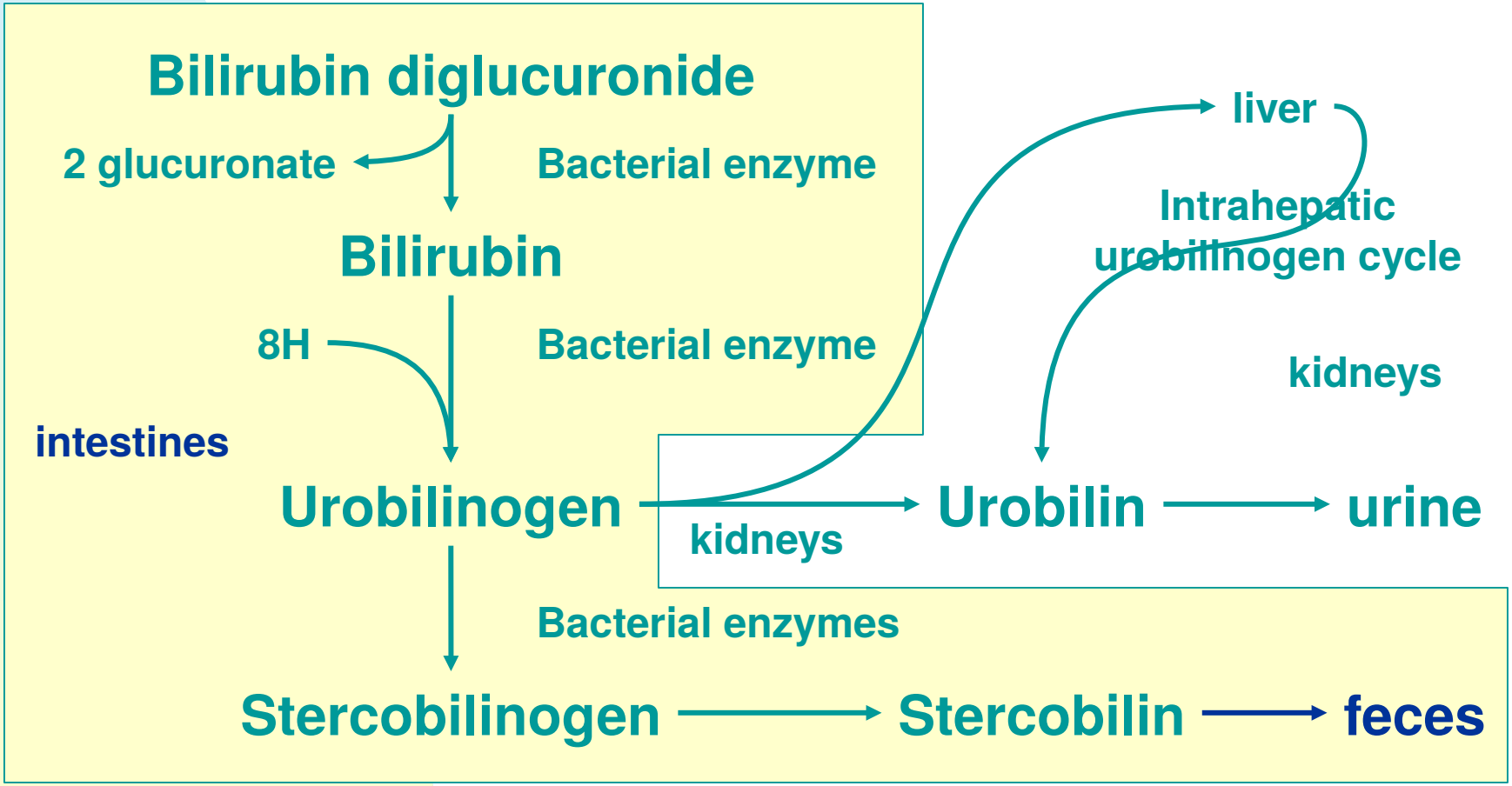
# Bilirubin Processing



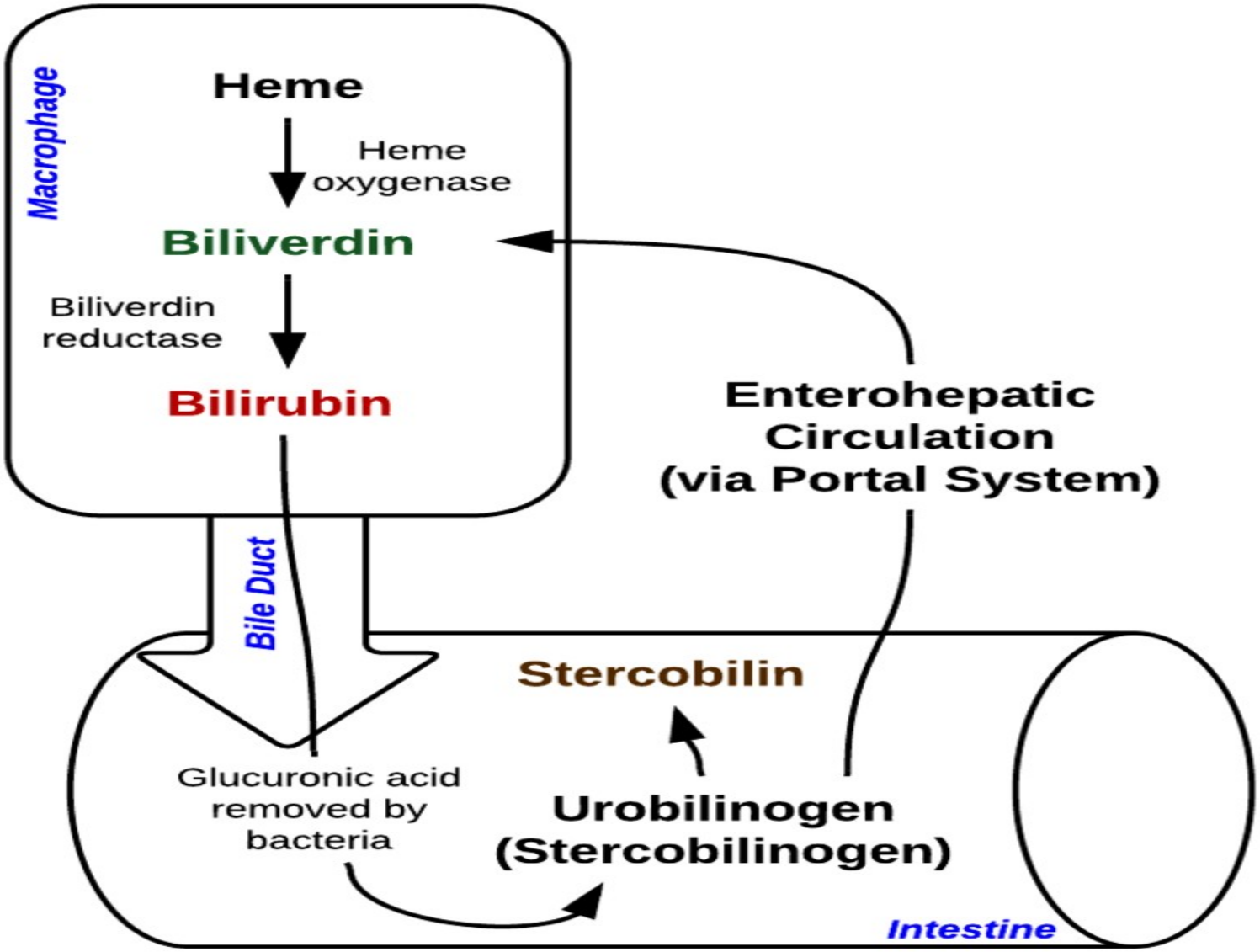
# structure of bilirubin diglucuronide



# Bilirubin Excretion



- Because bilirubin is highly insoluble in water, it must be converted into a soluble conjugate before elimination from the body.
- In the liver, uridine diphosphate (UDP)-glucuronyl transferase converts bilirubin to a mixture of monoglucuronides and diglucuronides, referred to as conjugated bilirubin, which is then secreted into the bile by an ATP-dependent transporter.
- This process is highly efficient under normal conditions, so plasma unconjugated bilirubin concentrations remain low.


Macrophage
Heme
Heme oxygenase
Biliverdin
Biliverdin reductase
Bilirubin
Enterohepatic Circulation (via Portal System)
Bile Duct
Stercobilin
Glucuronic acid removed by bacteria
Urobilinogen (Stercobilinogen)
Intestine

# Bilirubin Lab values

| Bilirubin form | Normal value |
|---|---|
| Total (elderly, adult, child) | 0.1 to 1.0 mg/dL |
| (newborn) | 1.0 to 12.0 mg/dL |
| Critical value (adult) | >12 mg/dL |
| (newborn) | >15 mg/dL |
| Pre-hepatic, unconjugated, indirect | 0.0 to 0.8 mg/dL |
| Post-hepatic, conjugated, direct | 0.0 to 0.25 mg/dL |
| Fecal urobilinogen | 40 to 280 mg/day |
| Urine | 0.0 to 0.02 mg/dL |

**Conjugated bilirubin - water soluble - direct reaction with dyes**

**Unconjugated bilirubin - water insoluble - alcohol is needed for dye (indirect) reaction**

# Difference of two bilirubins

| | free bilirubin | conjugated bilirubin |
|---|---|---|
| Binding with Glucuronic acid | no | yes |
| Reacting with the diazo reagent | Slow and indirect | Rapid and direct |
| solubility in water | small | large |
| Discharged via kidney | no | yes |
| Pass through the membrane of cell | yes | no |

# Hyperbilirubinemia

- Interferences at any one of the points of bilirubin processing described above can lead to a condition known as ***HYPERBILIRUBINEMIA***.

- As the name implies this disease is characterized by abnormally elevated levels of bilirubin in the blood.

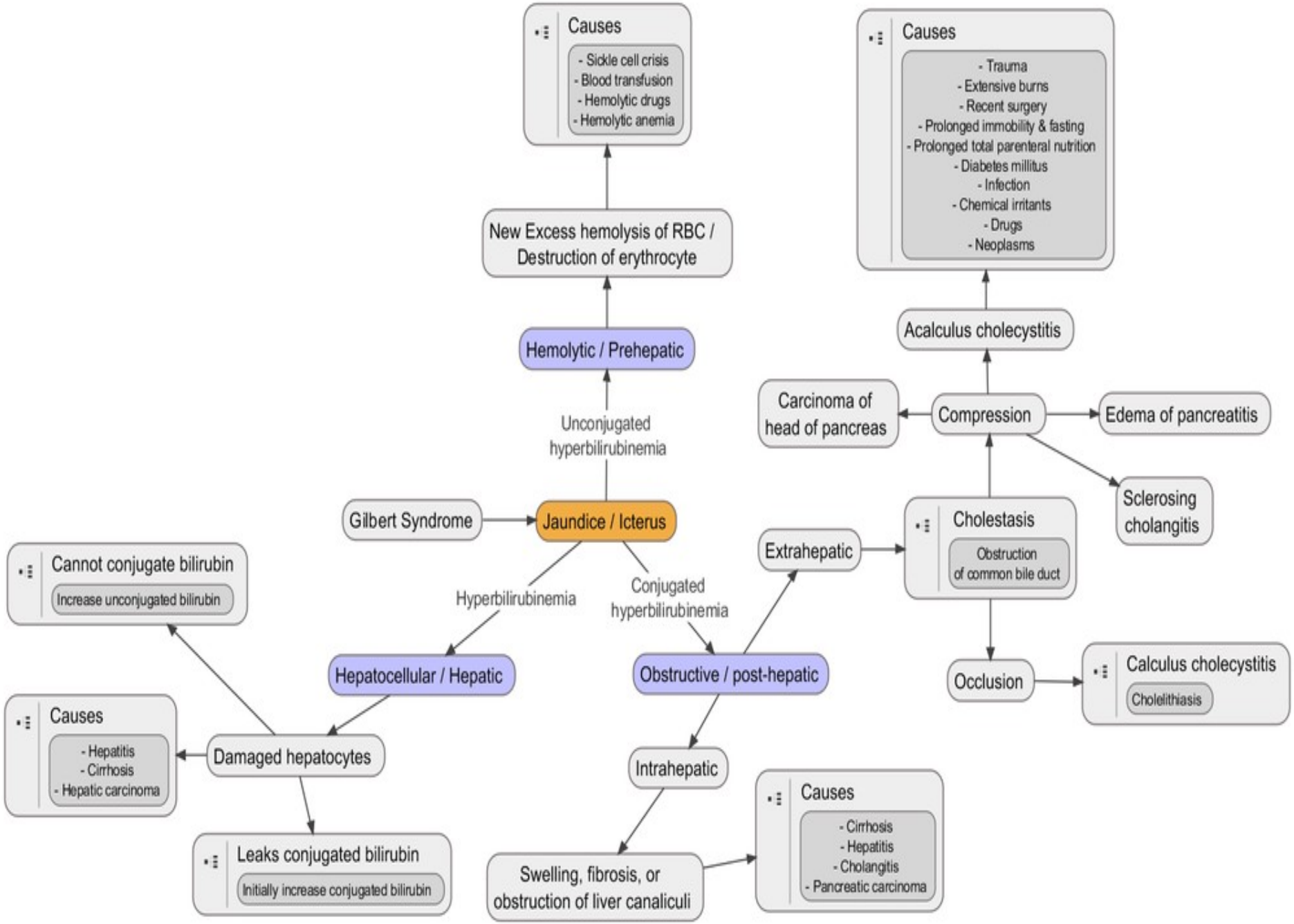
# SYMPTOMS

- Yellowing of the skin, scleras (white of the eye), and mucous membranes (jaundice)
- Detectable when total plasma bilirubin levels exceed 2mg/100mL



# Classification of Jaundice

- Jaundice is classified into three types.
  - ***1. Pre-hepatic/ hemolytic.***
  - ***2. Hepatic/ hepatocellular.***
  - ***3. Post-Hepatic/ cholestatic***


Causes
- Sickle cell crisis
- Blood transfusion
- Hemolytic drugs
- Hemolytic anemia
New Excess hemolysis of RBC / Destruction of erythrocyte
Hemolytic / Prehepatic
Unconjugated hyperbilirubinemia
Gilbert Syndrome
Jaundice / Icterus
Hyperbilirubinemia
Conjugated hyperbilirubinemia
Cannot conjugate bilirubin
Increase unconjugated bilirubin
Hepatocellular / Hepatic
Causes
- Hepatitis
- Cirrhosis
- Hepatic carcinoma
Damaged hepatocytes
Leaks conjugated bilirubin
Initially increase conjugated bilirubin
Obstructive / post-hepatic
Extrahepatic
Intrahepatic
Swelling, fibrosis, or obstruction of liver canaliculi
Causes
- Cirrhosis
- Hepatitis
- Cholangitis
- Pancreatic carcinoma
Cholestasis
Obstruction of common bile duct
Occlusion
Calculus cholecystitis
Cholelithiasis
Causes
- Trauma
- Extensive burns
- Recent surgery
- Prolonged immobility & fasting
- Prolonged total parenteral nutrition
- Diabetes millitus
- Infection
- Chemical irritants
- Drugs
- Neoplasms
Acalculus cholecystitis
Carcinoma of head of pancreas
Compression
Edema of pancreatitis
Sclerosing cholangitis

# Causes

1. **Increased bilirubin production**
2. **Reduced bilirubin uptake by hepatic cells**
3. **Disrupted intracellular conjugation**
4. **Disrupted secretion of bilirubin into bile canaliculi**
5. **Intra/extra-hepatic bile duct obstruction**

**Lead to increases in free (unconj.) bilirubin**

**Result in rise in conj. bilirubin levels**

# Classification of Jaundice

1. **Hemolytic Jaundice:** increased hemolysis of erythrocytes. Other causes may include sickle cell anemia, malaria, incompatible blood transfusion.

- Over production of bilirubin.
- Elevated levels of serum unconjugated bilirubin.
- More bilirubin is excreted into the bile – increased formation of urobilinogen & stercobilinogen.
- Dark brown color of feces (high stercobilinogen).

2. **Hepatic (hepatocellular) Jaundice:** damage to parenchymal cells caused dysfunction of liver.

- Major causes includes liver cirrhosis, viral hepatitis, hepatic damage associated with poisons or toxins.
- It adversely affects bilirubin uptake and its conjugation by liver cells.
- Elevated levels of conjugated and unconjugated bilirubin.
- Increase ALT & AST (released in circulation due to damage hepatocytes).

3. **Obstructive Jaundice:** due to the obstruction in the bile duct. It prevents the passage of bile into the intestine.

- It may be caused by gall stones, tumor, etc.
- In this case, the conjugated bilirubin from the liver enters the circulation.
- Elevated levels of serum conjugated bilirubin.
- Impairment in fat digestion & absorption (bile absence).
- Urine color is dark – elevated excretion of bilirubin.
- Clay color feces – absence of stercobilinogen.

| TYPES | IMPACT |
| --- | --- |
| ***Pre-hepatic/ hemolytic*** | Occurring prior to the hepatic area mainly due to 'Intrinsic defects in RB cells' OR 'Extrinsic causes external to RB cells' |
| ***Hepatic/ hepatocellular*** | Located within the hepatic area, due to disease of parenchymal cells of liver. |
| ***Post-Hepatic/ cholestatic*** | Located after the conjugation of bilirubin in the hepatic area caused due to obstruction of biliary passage. |

# Laboratory results in patients with jaundice

| | normal | Hemolytic jaundice | Hepatocellular jaundice | Obstructive jaundice |
|---|---|---|---|---|
| Serum bilirubin | | | | |
| total | < 1mg/dl | > 1mg/dl | > 1mg/dl | > 1mg/dl |
| direct | 0~ 0.8mg/dl | | ↑ | ↑↑ |
| indirect | < 1 | ↑↑ | ↑ | |
| Urine bile pigments | | | | |
| urobilirubin | – | – | + + | + + |
| urobilinogen | A few | ↑ | uncertainty | ↓ |
| urobilin | A few | ↑ | uncertainty | ↓ |
| Color of feces | normal | dark | Simple or normal | Clay color |

**Neonatal jaundice:** Hyperbilirubinaemia accumulation of bilirubin (unconjugated) in certain brain regions.

Resulted in hyperbilirubinemic toxic encephalopathy or ***kernicterus***, that causes mental retardation, manifesting as various neurological deficits, seizures, abnormal reflexes and eye movements.

# DISRUPTED INTRACELLULAR CONJUGATION (unconj. Hyperbilirubinemia)

- Crigler-Najjar Syndrome, Type I (CN-I) : congenital non-hemolytic jaundice. Recessive allele; mutation-induced loss of conjugating ability in the critical hepatic enzyme ***glucuronosyltransferase***
- CN-II : rare hereditary disorder (due to severe defect in bilirubin conjugation).

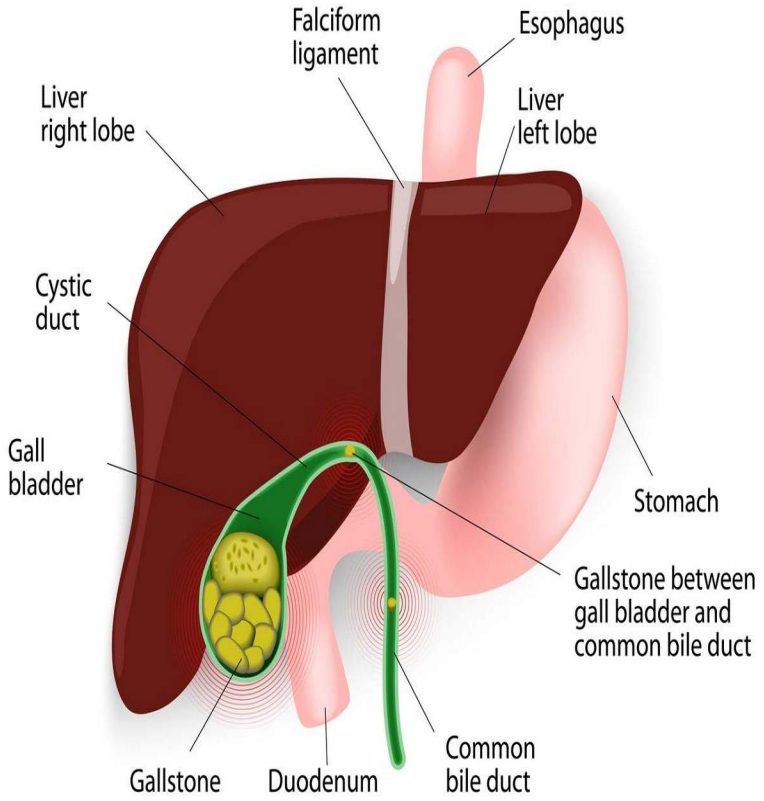
Greatly reduced but detectable glucuronosyltransferase activity due to mutation (predominantly recessive); enzymatic activity can be induced by drugs

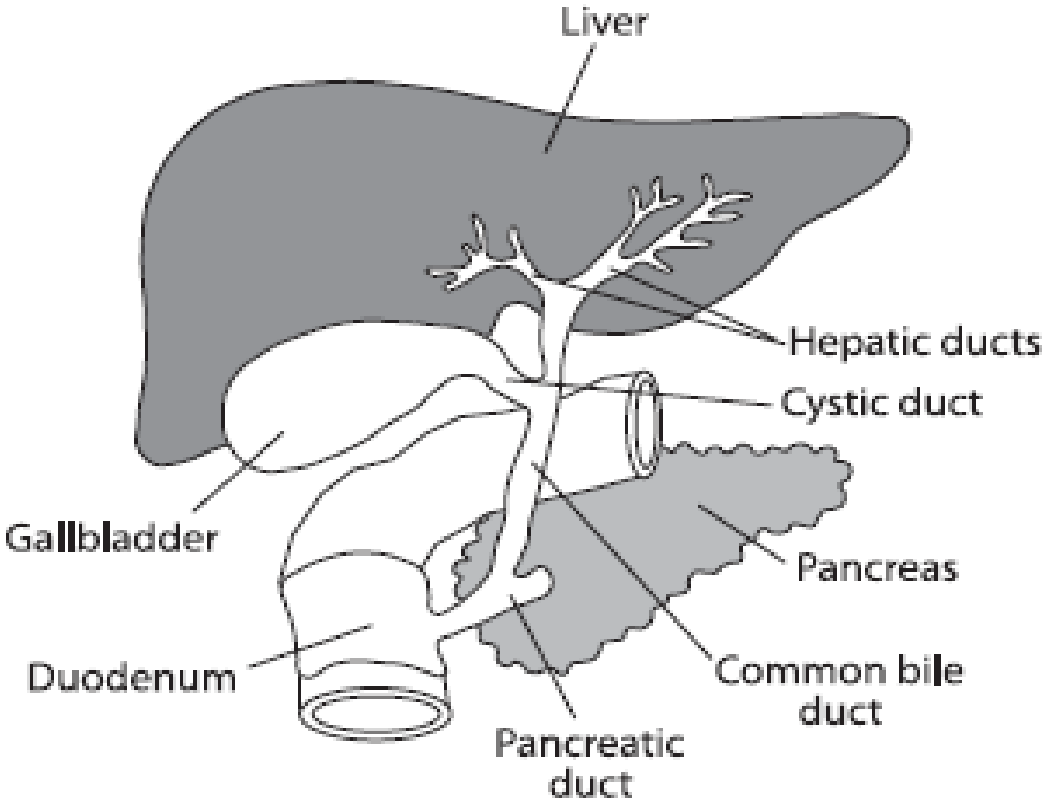
- Gilbert's Syndrome (elevated Unconjugated bilirubin)

***glucuronosyl transferase*** activity reduced to 10-30% of normal; also accompanied by defective bilirubin uptake mechanism

# Treatment / Management

- Bile duct blockage – *Gallstones, cancer, pancreatitis.*



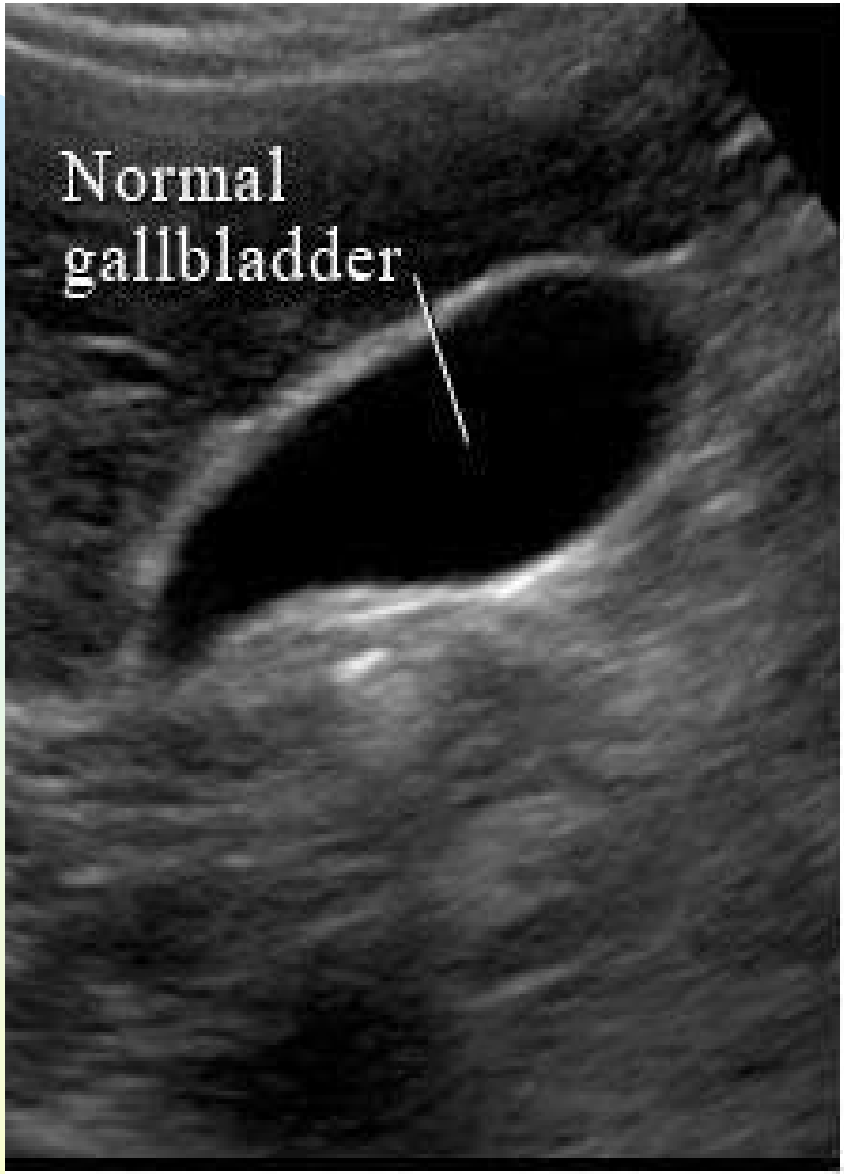



Figure 1




Figure 2

- The most common treatment is surgery to remove the gallbladder.
- This technique (operation) is called a ***cholecystectomy.***
- Management may involve treating infectious causes.
- Stopping medication that could be contributing factor for liver damage.
- Neonatal - may be treated with phototherapy or exchanged transfusion.

# Thank you